

Merkle Tree Certs

Why, What, and When

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Let's Encrypt

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WHY

(do we need them?)

Post-Quantum Certificate Sizes

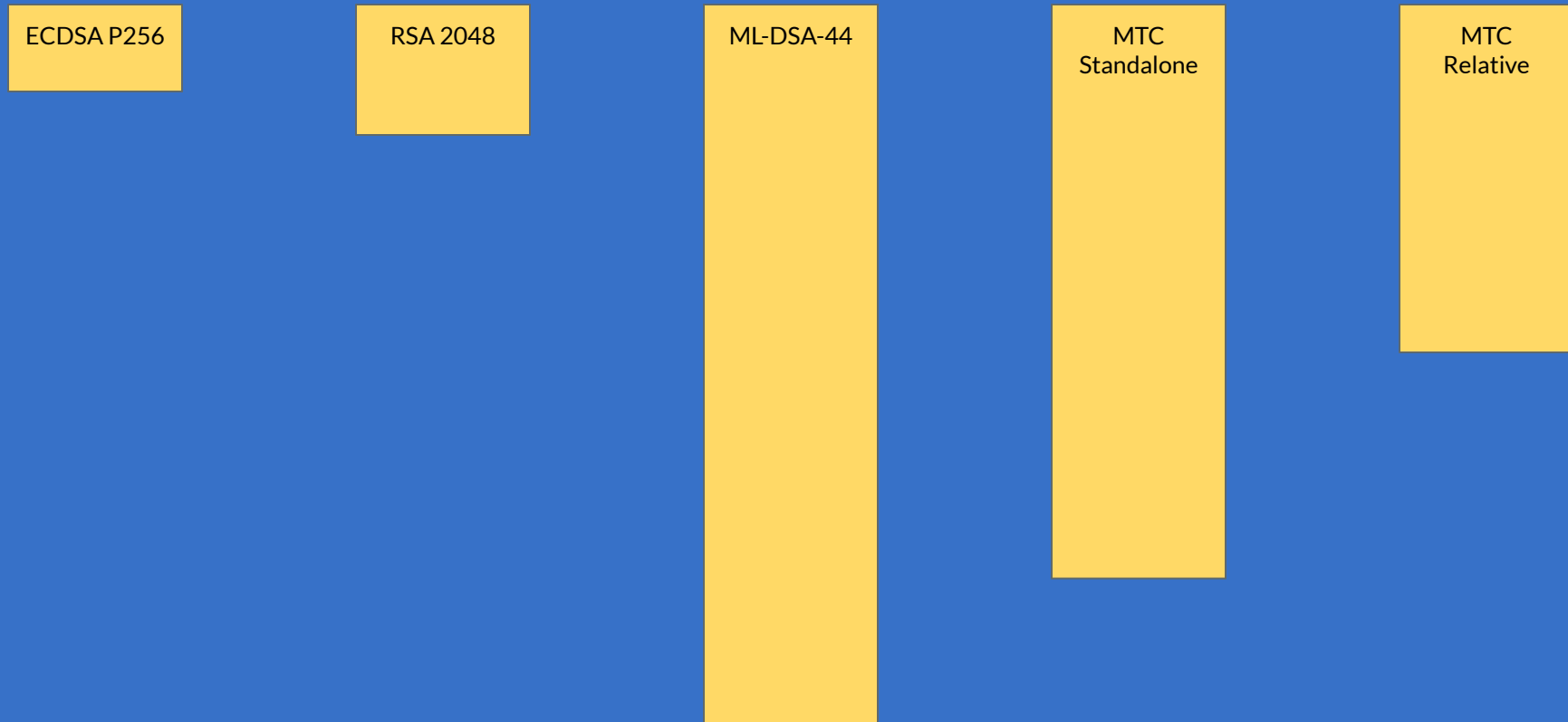
ECDSA P256

RSA 2048

ML-DSA-44

MTC
Standalone

MTC
Relative

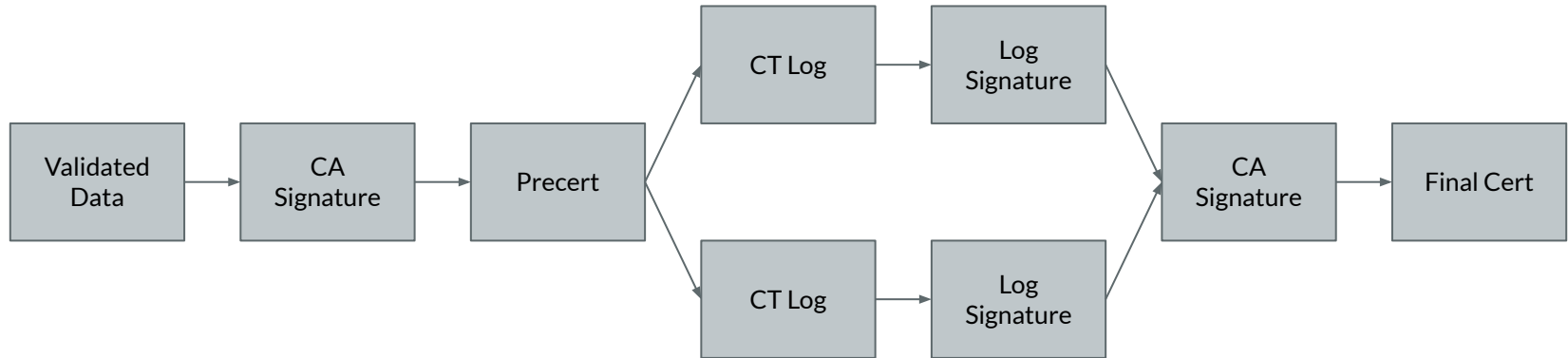




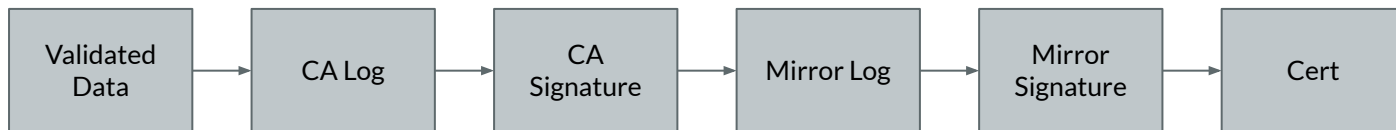
WHAT

(even are they?)

Don't log what you issue, issue by logging



Don't log what you issue, issue by logging



Transparency Is Built In

Certificate:

```
Data:
  Version: 3 (0x2)
  Serial Number:
    05:18:ab:24:c4:8c:ac:79:c3:80:7e:8b:31:2b:f4:e1:9c:10
  Yadda yadda yadda
  X509v3 extensions:
    CT Precertificate SCTs:
      Signed Certificate Timestamp:
        Version : v1 (0x0)
        Log ID   : 46:AF:86:3D:3B:3E:E5:9F:A5:77:DE:A8:24:5D:36:B0:
                     D9:ED:22:A2:23:F4:61:77:41:22:94:52:EE:95:50:5F
        Timestamp : May 7 17:13:07.930 2026 GMT
        Extensions: 00:00:05:00:05:F4:2C:2D
        Signature : ecdsa-with-SHA256
                     30:44:02:20:70:A7:37:61:13:57:04:08:1A:5C:9A:9A:
                     4A:A1:D9:B2:09:E0:87:12:6F:07:0D:F7:B3:2F:0F:79:
                     52:BF:3C:74:02:20:36:6F:3A:4A:35:0F:B8:75:6B:D6:
                     41:32:79:7E:2B:EC:48:69:F9:0A:6F:F3:51:AA:F7:79:
                     EC:07:B8:6F:8A:9E
      Signed Certificate Timestamp:
        Version : v1 (0x0)
        Log ID   : AF:67:88:3B:57:B0:4E:DD:8F:A6:D9:7E:F6:2E:A8:EB:
                     81:0A:C7:71:60:F0:24:5E:55:D6:0C:2F:E7:85:87:3A
        Timestamp : May 7 17:13:08.116 2026 GMT
        Extensions: none
        Signature : ecdsa-with-SHA256
                     30:45:02:21:00:B9:22:0F:40:8F:FA:31:8F:EE:F6:0F:
                     49:AA:1C:2F:41:0E:0B:BD:FD:31:B6:61:CC:AB:0B:F9:
                     4B:63:45:BC:82:02:20:76:77:9A:24:96:1D:B5:F6:47:
                     1B:EF:9E:62:4F:FE:B7:D6:E7:F6:0E:E6:6C:74:3D:03:
                     CA:E1:F5:CA:2E:5F:E9
  Signature Algorithm: ecdsa-with-SHA384
  Signature Value:
    30:65:02:30:28:f2:37:63:3f:b0:7a:58:d4:5f:8d:7b:ca:b3:
    44:cf:cb:93:41:8f:95:ac:44:88:cb:ba:2a:cf:9d:77:49:94:
    f7:bc:e1:ae:f1:58:62:f8:12:b5:12:5c:ba:1e:d6:a1:02:31:
    00:e0:13:ec:a5:86:12:a2:4a:35:ba:0d:94:95:20:33:50:f4:
    86:41:55:f5:1f:1c:7a:68:4e:0a:85:f2:ac:6b:54:1f:1a:d3:
    f3:d2:cf:b2:b9:4c:39:03:40:ff:c1:7d:e7
```

"I certify that this cert has been durably logged..."

"I certify the contents of this cert have been validated..."

Transparency Is Built In

Certificate:

Data:

Version: 3 (0x2)

Serial Number:

05:18:ab:24:c4:8c:ac:79:c3:80:7e:8b:31:2b:f4:e1:9c:10

Yadda yadda yadda

Signature Algorithm: 1.3.6.1.5.5.7.6.X (id-alg-mtcProof)

Signature Value:

c2:8e:c5:2d:dc:6c:38:97:fd:93:47:84:0a:c6:51:7e:39:6c:
92:03:5a:3e:99:b2:f1:d9:45:3b:d7:f7:c5:79:a8:9c:dd:51:
98:59:c6:07:8a:89:15:80:c5:d8:7b:39:a3:dd:21:50:2e:9a:
1d:3b:1d:99:89:bc:dd:7f:b1:37:21:8c:5b:ee:33:98:8e:5b:
52:56:06:af:e5:d2:70:02:35:12:df:49:5c:24:36:44:67:da:
1a:54:47:62:5c:b5:50:3b:08:71:11:f0:8e:d4:21:32:86:ba:
bb:f6:cc:fd:fa:96:65:d4:6d:fe:10:36:92:d4:1e:77:ca:91:
a3:e4:ea:38:82:ac:5b:6c:97:1d:d8:aa:56:ea:24:c8:6e:18:
61:66:00:56:9b:29:39:bb:f1:f1:14:1b:e1:1f:22:6f:f0:e0:
b2:14:1e:82:e6:98:3e:49:f3:95:cb:ed:7b:d9:2f:72:1e:2a:
03:73:65:20:50:3c:03:68:aa:e6:58:d7:5a:e9:60:3e:a7:9b:
94:64:62:ff:0c:7e:ba:b8:a5:1c:66:72:74:96:4d:57:d4:21:
43:f7:b9:05:14:27:f3:b1:19:ad:55:6c:da:eb:0e:aa:33:77:
4f:ea:07:5a:b9:b1:9c:d0:4b:2c:c3:59:74:53:07:80:07:17:
28:8d:60:96:b3:5d:86:1d:93:79:ae:b9:5a:55:e6:71:6f:d4:
40:ba:9f:3f:87:20:b8:4f:dc:74:28:91:cb:47:88:9d:3a:92:
29:4d:e4:9b:74:a8:7e:f8:5f:ac:a9:62:24:22:78:a7:9a:a7:
a9:f9:66:13:62:14:1f:f1:96:0b:a7:1e:47:1d:53:78:84:70:
f8:65:a2:80:fd:10:1e:0f:f6:59:aa:3d:e5:58:f7:b9:7b:dc:
09:17:43:b1:95:e3:fa:08:13:6d:9f:f2:eb:4b:89:2f:92:cd:
69:af:a6:f6:fd:c3:4b:45:2d:ab:18:d5:ca:ef:15:0a:de:52:
d6:45:87:1d:f4:7e:90:77:23:29:6b:98:47:c5:35:30:70:a3:
4b:a4:4d:89:c0:d3:1f:9d:83:84:c3:ef:3d:0d:b4:19:42:13:
dd:91:d0:3a:ea:8e:91:83:c6:61:7d:1f:e2:ee:42:a5:fc:e2:
3f:71:c1:a7:c6:98:c0:32:64:42:93:d1:16:72:50:23:c2:ab:
c1:15:8f:43:30:80:9f:89:a9:b6:21:da:98:2b:4b:e8:1f:65:
7c:14:49:ab:d1:19:a2:89:41:6e:17:57:0d:22:c4:ef:fd:f4:
f2:3b:ef:5b:3d:54:a1:dc:e3:4f:f6:0c:4a:7d:88:9d:2b:a2:
1b:06:09:27:e8:a2:6d:d1:8d:70:66:61:42:6f:5e:f2:04:ce:
df:49:1f:6f:5a:35:bc:b5:99:00:41:5e:27:93:dc:3a:e0:18:
bf:8e:23:22:bd:f9:5a:2d:45:6a:40:76:69:62:ea:55:5b:81:
e8:ce:cd:bc:d0:3e:40:79:ba:08:59:ef:12:13:8b:e8:7f:23:

"I certify that this cert has been durably logged..."

"I certify the contents of this cert have been validated *and* that it has been durably logged..."

Just A Funny Signature Algorithm

Signature Algorithm: `ecdsa-with-SHA256`

Signature Value: `ecdsa(hash(data))`

Just A Funny Signature Algorithm

Signature Algorithm: id-alg-mtcProof

Signature Value:

```
struct {  
    inclusionProof []merkleHash  
    signatures []struct {  
        logID OID  
        signature mldsa(top of inclusion proof)  
    }  
}
```

The Landmark Optimization

Signature Algorithm: `id-alg-mtcProof`

Signature Value:

```
struct {  
    inclusionProof []merkleHash  
}
```

Out-of-band: ship landmark signatures to the browser



WHEN

(can I have one?)



Very Soon!

IETF has chartered PLANTS WG

IETF PLANTS WG has adopted MTC draft

C2SP has specified tlog-mirror and tlog-cosignature

Chrome has a draft MTC root program policy

Very Soon!

Chrome has deployed validation of MTCAs behind a flag

Let's Encrypt is working on an ACME MTCA implementation

Geomys is working on a mirror implementation

Cloudflare and several CAs are working on MTCAs, too



DEMO TIME



HOW

(can I help?)

Contribute to Standardization

IETF WG: <https://datatracker.ietf.org/wg/plants/about/>

MTCs: <https://github.com/ietf-plants-wg/merkle-tree-certs>

C2SP tlog: <https://github.com/C2SP/C2SP>



Questions?